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Inventor(s)	Huang; Chin-Ling

Semiconductor device structure with fuse and resistor and method for preparing the same

Abstract

A semiconductor device structure includes an isolation structure disposed in a semiconductor substrate. The semiconductor device structure also includes a fuse and a resistor electrode disposed in the semiconductor substrate. The isolation structure is disposed between the fuse and the resistor electrode, and the isolation structure is closer to the resistor electrode than the fuse. The semiconductor device structure further includes a source/drain (S/D) region disposed in the semiconductor substrate and between the fuse and the isolation structure. The S/D region is electrically connected to the resistor electrode.

Inventors:	Huang; Chin-Ling (Taoyuan, TW)
Applicant:	NANYA TECHNOLOGY CORPORATION (New Taipei, TW)
Family ID:	93265647
Assignee:	NANYA TECHNOLOGY CORPORATION (New Taipei, TW)
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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8598679	12/2012	Cheng	257/E23.149	H10D 89/911
2011/0303982	12/2010	Chung et al.	N/A	N/A
2012/0112311	12/2011	Cho et al.	N/A	N/A
2012/0196423	12/2011	Kanike et al.	N/A	N/A
2014/0183657	12/2013	Lim et al.	N/A	N/A
2017/0170116	12/2016	He et al.	N/A	N/A
2019/0067191	12/2018	Wong et al.	N/A	N/A

OTHER PUBLICATIONS

Office Action and and the search report mailed on Jul. 29, 2024 related to Taiwanese Application No. 113108411. cited by applicant

Office Action and and the search report mailed on Mar. 26, 2024 related to Taiwanese Application No. 112125795. cited by applicant

Primary Examiner: Ho; Tu-Tu V

Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a semiconductor device structure and a method for preparing the same, and more particularly, to a semiconductor device structure with a fuse and a resistor and a method for preparing the same.

DISCUSSION OF THE BACKGROUND

(2) Semiconductor devices are essential for many modern applications. With the advancement of electronic technology, semiconductor devices are becoming smaller in size while providing greater functionality and including greater amounts of integrated circuitry. Due to the miniaturized scale of semiconductor devices, various types and dimensions of semiconductor devices providing different functionalities are integrated and packaged into a single module. Furthermore, numerous manufacturing operations are implemented for integration of various types of semiconductor devices.

(3) However, the manufacturing and integration of semiconductor devices involve many complicated steps and operations. Integration in semiconductor devices becomes increasingly complicated. An increase in complexity of manufacturing and integration of the semiconductor device may cause problems. For example, a shrinking resistor formed by a conventional process

flow may have an insufficient sheet resistance. Accordingly, there is a continuous need to improve the manufacturing process of semiconductor devices so that the problems can be addressed.

(4) This Discussion of the Background section is provided for background information only. The statements in this Discussion of the Background are not an admission that the subject matter disclosed in this section constitutes prior art to the present disclosure, and no part of this Discussion of the Background section may be used as an admission that any part of this application, including this Discussion of the Background section, constitutes prior art to the present disclosure.

SUMMARY

(5) In one embodiment of the present disclosure, a semiconductor device structure is provided. The semiconductor device structure includes an isolation structure disposed in a semiconductor substrate. The semiconductor device structure also includes a fuse and a resistor electrode disposed in the semiconductor substrate. The isolation structure is disposed between the fuse and the resistor electrode, and the isolation structure is closer to the resistor electrode than the fuse. The semiconductor device structure further includes a source/drain (S/D) region disposed in the semiconductor substrate and between the fuse and the isolation structure. The S/D region is electrically connected to the resistor electrode.

(6) In an embodiment, an air gap is enclosed in the fuse. In an embodiment, the fuse is an electrical fuse. In an embodiment, a top width of the resistor electrode is greater than a top width of the fuse. In an embodiment, the fuse and the resistor electrode comprise polysilicon. In an embodiment, a dopant concentration of the fuse is greater than a dopant concentration of the resistor electrode. In an embodiment, a top surface of the fuse is substantially coplanar with a top surface of the resistor electrode.

(7) In an embodiment, the semiconductor device structure further includes a well region disposed in the semiconductor substrate, wherein the resistor electrode is disposed over the well region. In an embodiment, the well region adjoins the isolation structure. In an embodiment, the semiconductor device structure further includes a dielectric layer disposed over the semiconductor substrate, wherein a first portion of the dielectric layer extends between the fuse and the semiconductor substrate. In an embodiment, a second portion of the dielectric layer extends between the resistor electrode and the semiconductor substrate. In an embodiment, the isolation structure is covered by the dielectric layer.

(8) In an embodiment, the semiconductor device structure further includes an interlayer-dielectric (ILD) layer disposed over the semiconductor substrate, wherein the fuse and the resistor electrode are covered by the ILD layer. In an embodiment, the semiconductor device structure further includes an interconnect structure disposed in the ILD layer, wherein the S/D region is electrically connected to the resistor electrode through the interconnect structure.

(9) In an embodiment, the interconnect structure includes a first conductive via and a second conductive via disposed in the ILD layer, wherein the first conductive via is electrically connected to the S/D region, and the second conductive via is electrically connected to the resistor electrode. In addition, the interconnect structure also includes a conductive layer disposed over the ILD layer and electrically connecting the first conductive via and the second conductive via. In an embodiment, the semiconductor device structure further includes a conductive structure disposed in the semiconductor structure and over the isolation structure, wherein the S/D region is electrically connected to the resistor electrode through the conductive structure.

(10) In another embodiment of the present disclosure, a semiconductor device structure is provided. The semiconductor device structure includes a fuse and a resistor electrode disposed in a semiconductor substrate. A dopant concentration of the fuse is greater than a dopant concentration of the resistor electrode. The semiconductor device structure also includes an isolation structure disposed in the semiconductor substrate and between the fuse and the resistor electrode. The semiconductor device structure further includes a source/drain (S/D) region disposed in the semiconductor substrate and between the fuse and the isolation structure. The S/D region is

electrically connected to the resistor electrode.

(11) In an embodiment, an air gap is enclosed in the fuse. In an embodiment, the fuse is an electrical fuse. In an embodiment, a top width of the resistor electrode is greater than a top width of the fuse. In an embodiment, the isolation structure is closer to the resistor electrode than the fuse. In an embodiment, the fuse and the resistor electrode comprise polysilicon.

(12) In an embodiment, the semiconductor device structure further includes a well region disposed in the semiconductor substrate, wherein the resistor electrode is disposed over the well region, and a conductivity type of the well region is the same as a conductivity type of the S/D region. In an embodiment, the semiconductor device structure further includes a dielectric layer disposed over the semiconductor substrate, wherein the fuse is separated from the semiconductor substrate by a first portion of the dielectric layer, and the resistor electrode is separated from the well region by a second portion of the dielectric layer.

(13) In an embodiment, a top surface and a sidewall of the isolation structure are in direct contact with the dielectric layer. In an embodiment, the semiconductor device structure further includes a conductive structure disposed in the semiconductor substrate and over the isolation structure, wherein the conductive structure is in direct contact with the S/D region and the resistor electrode. In an embodiment, a top surface of the fuse, a top surface of the resistor electrode, and a top surface of the conductive structure are substantially coplanar with each other.

(14) In yet another embodiment of the present disclosure, a method for preparing a semiconductor device structure is provided. The method includes forming an isolation structure in a semiconductor substrate, and recessing the semiconductor substrate to form a first opening and a second opening. The first opening and the second opening are on opposite sides of the isolation structure, and a width of the second opening is greater than a width of the first opening. The method also includes forming an electrode layer over the semiconductor substrate, and polishing the electrode layer to form a fuse in the first opening and a resistor electrode in the second opening. The method further includes forming a source/drain (S/D) region in the semiconductor substrate. The S/D region is between the fuse and the isolation structure.

(15) In an embodiment, an air gap is enclosed in the fuse. In an embodiment, the method further includes forming a well region in the semiconductor substrate before the first opening and the second opening are formed, wherein the second opening is formed by removing an upper portion of the well region. In an embodiment, a bottom surface of the well region is higher than a bottom surface of the isolation structure. In an embodiment, the isolation structure is closer to the second opening than the first opening. In an embodiment, a sidewall of the isolation structure is exposed by the second opening before the electrode layer is formed. In an embodiment, the method further includes depositing a dielectric layer over the semiconductor substrate before the electrode layer is formed, wherein the dielectric layer extends into the first opening and the second opening. In an embodiment, the dielectric layer is in direct contact with a sidewall of the isolation structure. In an embodiment, the method further includes performing an ion implantation process on the electrode layer before the electrode layer is polished.

(16) In an embodiment, a top surface of the fuse is substantially coplanar with a top surface of the resistor electrode after the electrode layer is polished. In an embodiment, the method further includes forming an interlayer-dielectric (ILD) layer over the semiconductor substrate, and forming an interconnect structure in the ILD layer, wherein the S/D region is electrically connected to the resistor electrode through the interconnect structure. In an embodiment, the forming the interconnect structure includes etching the ILD layer to form a third opening exposing the S/D region and a fourth opening exposing the resistor electrode, forming a first conductive via in the third opening and a second conductive via in the fourth opening, and forming a conductive layer over the ILD layer and electrically connecting the first conductive via and the second conductive via. In an embodiment, the method further includes etching an upper portion of the isolation structure to form a fifth opening after the electrode layer is polished, and forming a conductive

structure in the fifth opening before the S/D region is formed, wherein the conductive structure is used to electrically connect the resistor electrode and the S/D region.

(17) Embodiments of a semiconductor device structure are provided in the disclosure. The semiconductor device structure includes a transistor and a resistor connected in series and formed by an integrated process flow. In particular, a gate electrode of the transistor and a resistor electrode of the resistor are formed in a semiconductor substrate by the same process steps. Therefore, the resistor may have high sheet resistance without using additional masks or process steps. As a result, the associated costs may be reduced, and the performance of the semiconductor device structure may be improved.

(18) Moreover, the semiconductor device structure includes a fuse and a resistor connected in series and formed by an integrated process flow. Since the fuse and a resistor electrode of the resistor are formed in a semiconductor substrate by the same process steps, the associated costs may be reduced, and the performance of the semiconductor device structure may be improved.

(19) The foregoing has outlined rather broadly the features and technical advantages of the present disclosure in order that the detailed description of the disclosure that follows may be better understood. Additional features and advantages of the disclosure will be described hereinafter, and form the subject of the claims of the disclosure. It should be appreciated by those skilled in the art that the conception and specific embodiment disclosed may be readily utilized as a basis for modifying or designing other structures or processes for carrying out the same purposes of the present disclosure. It should also be realized by those skilled in the art that such equivalent constructions do not depart from the spirit and scope of the disclosure as set forth in the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It should be noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 is a cross-sectional view illustrating a semiconductor device structure, in accordance with some embodiments.

(3) FIG. 2 is a circuit diagram of the semiconductor device structure in FIG. 1.

(4) FIG. 3 is a cross-sectional view illustrating a modified semiconductor device structure, in accordance with some embodiments.

(5) FIG. 4 is a flow diagram illustrating a method of forming a semiconductor device structure, in accordance with some embodiments.

(6) FIG. 5 is a cross-sectional view illustrating an intermediate stage of forming isolation structures and a well region during the formation of the semiconductor device structure, in accordance with some embodiments.

(7) FIG. 6 is a cross-sectional view illustrating an intermediate stage of forming openings between the isolation structures during the formation of the semiconductor device structure, in accordance with some embodiments.

(8) FIG. 7 is a cross-sectional view illustrating an intermediate stage of forming a dielectric layer during the formation of the semiconductor device structure, in accordance with some embodiments.

(9) FIG. 8 is a cross-sectional view illustrating an intermediate stage of forming an electrode layer during the formation of the semiconductor device structure, in accordance with some embodiments.

(10) FIG. 9 is a cross-sectional view illustrating an intermediate stage of performing an ion implantation process on the electrode layer during the formation of the semiconductor device

structure, in accordance with some embodiments.

(11) FIG. **10** is a cross-sectional view illustrating an intermediate stage of performing an ion implantation process on the electrode layer during the formation of the semiconductor device structure, in accordance with some embodiments.

(12) FIG. **11** is a cross-sectional view illustrating an intermediate stage of polishing the electrode layer during the formation of the semiconductor device structure, in accordance with some embodiments.

(13) FIG. **12** is a cross-sectional view illustrating an intermediate stage of forming source/drain regions during the formation of the semiconductor device structure, in accordance with some embodiments.

(14) FIG. **13** is a cross-sectional view illustrating an intermediate stage of forming an interlayer-dielectric layer during the formation of the semiconductor device structure, in accordance with some embodiments.

(15) FIG. **14** is a cross-sectional view illustrating an intermediate stage of forming a patterned mask over the interlayer-dielectric layer during the formation of the semiconductor device structure, in accordance with some embodiments.

(16) FIG. **15** is a cross-sectional view illustrating an intermediate stage of partially removing the interlayer-dielectric layer and the dielectric layer during the formation of the semiconductor device structure, in accordance with some embodiments.

(17) FIG. **16** is a cross-sectional view illustrating an intermediate stage of performing an ion implantation process on the electrode layer during the formation of the semiconductor device structure, in accordance with some embodiments.

(18) FIG. **17** is a cross-sectional view illustrating an intermediate stage of performing an ion implantation process on the electrode layer during the formation of the semiconductor device structure, in accordance with some embodiments.

(19) FIG. **18** is a cross-sectional view illustrating an intermediate stage of forming an opening in the semiconductor substrate during the formation of the semiconductor device structure, in accordance with some embodiments.

(20) FIG. **19** is a cross-sectional view illustrating an intermediate stage of forming a conductive layer during the formation of the semiconductor device structure, in accordance with some embodiments.

(21) FIG. **20** is a cross-sectional view illustrating an intermediate stage of forming a conductive structure during the formation of the semiconductor device structure, in accordance with some embodiments.

(22) FIG. **21** is a cross-sectional view illustrating an intermediate stage of forming an electrode layer during the formation of the semiconductor device structure, in accordance with some embodiments.

(23) FIG. **22** is a cross-sectional view illustrating an intermediate stage of performing an ion implantation process on the electrode layer during the formation of the semiconductor device structure, in accordance with some embodiments.

(24) FIG. **23** is a cross-sectional view illustrating an intermediate stage of performing an ion implantation process on the electrode layer during the formation of the semiconductor device structure, in accordance with some embodiments.

(25) FIG. **24** is a cross-sectional view illustrating an intermediate stage of polishing the electrode layer during the formation of the semiconductor device structure, in accordance with some embodiments.

(26) FIG. **25** is a cross-sectional view illustrating a semiconductor device structure, in accordance with some embodiments.

(27) FIG. **26** is a cross-sectional view illustrating an intermediate stage of forming an opening in the semiconductor substrate during the formation of the semiconductor device structure, in

accordance with some embodiments.

(28) FIG. 27 is a cross-sectional view illustrating an intermediate stage of forming a conductive layer during the formation of the semiconductor device structure, in accordance with some embodiments.

(29) FIG. 28 is a cross-sectional view illustrating an intermediate stage of forming a conductive structure during the formation of the semiconductor device structure, in accordance with some embodiments.

(30) FIG. 29 is a cross-sectional view illustrating a modified semiconductor device structure, in accordance with some embodiments.

DETAILED DESCRIPTION

(31) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(32) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(33) FIG. 1 is a cross-sectional view illustrating a semiconductor device structure **300a**, in accordance with some embodiments. As shown in FIG. 1, the semiconductor device structure **300a** includes a transistor **100** and a resistor **200**. In the semiconductor device structure **300a**, a plurality of isolation structures **105a**, **105b** and **105c** are disposed in a semiconductor substrate **101**, the active region of the transistor **100** is defined by the isolation structures **105a** and **105b**, and the active region of the resistor **200** is defined by the isolation structures **105b** and **105c**, in accordance with some embodiments. It should be noted that the number of isolation structures per semiconductor device structure is not limited to three, and may be less or more.

(34) In some embodiments, the semiconductor device structure **300a** also includes a gate electrode **115a** and source/drain (S/D) regions **121a**, **121b** in the active region of the transistor **100** (i.e., between the isolation structures **105a** and **105b**). The gate electrode **115a** is located between the S/D regions **121a** and **121b**, and the gate electrode **115a** and the S/D regions **121a**, **121b** are disposed in the semiconductor substrate **101**.

(35) In some embodiments, the semiconductor device structure **300a** further includes a well region **103** and a resistor electrode **115b** in the active region of the resistor **200** (i.e., between the isolation structures **105b** and **105c**). The well region **103** and the resistor electrode **115b** are disposed in the semiconductor substrate **101**, and the resistor electrode **115b** is disposed over the well region **103**. In addition, the well region **103** adjoins the isolation structures **105b** and **105c**. It should be noted that the isolation structure **105b** between the active regions of the transistor **100** and the resistor **200** is closer to the resistor electrode **115b** than the gate electrode **115a**, in accordance with some embodiments.

(36) Moreover, the semiconductor device structure **300a** includes a dielectric layer **113'** disposed

over the semiconductor substrate **101**, in accordance with some embodiments. In particular, the dielectric layer **113'** has a first portion and a second portion below the top surface **101T** of the semiconductor substrate **101**, the first portion is between the gate electrode **115a** and the semiconductor substrate **101**, and the second portion is between the resistor electrode **115b** and the semiconductor substrate **101**. In some embodiments, the gate electrode **115a** is separated from the semiconductor substrate **101** by the first portion of the dielectric layer **113'**, and the resistor electrode **115b** is separated from the well region **103** in the semiconductor substrate **101** by the second portion of the dielectric layer **113'**.

(37) In addition, in some embodiments, the resistor electrode **115b** is separated from the isolation structures **105b** and **105c** by the second portion of the dielectric layer **113'**. In some embodiments, the isolation structures **105a**, **105b** and **105c** are covered by the dielectric layer **113'**, and the S/D regions **121a** and **121b** are partially covered by the dielectric layer **113'**.

(38) Still referring to FIG. **1**, the semiconductor device structure **300a** includes an interlayer-dielectric (ILD) layer **123'** disposed over the dielectric layer **113'**, and an interconnect structure **137** disposed over the ILD layer **123'**, in accordance with some embodiments. More specifically, the interconnect structure **137** includes a plurality of conductive vias **133a**, **133b**, **133c** and **133d**, and a plurality of conductive layers **135a**, **135b** and **135c**.

(39) In some embodiments, the conductive vias **133a**, **133b**, **133c** and **133d** are disposed in the ILD layer **123'**, and the conductive layers **135a**, **135b** and **135c** are disposed over the ILD layer **123'**. In some embodiments, the conductive vias **133a** and **133b** penetrate through the ILD layer **123'** and the dielectric layer **113'** to electrically connect the S/D regions **121a** and **121b**, and the conductive vias **133c** and **133d** penetrate through the ILD layer **123'** to electrically connect the resistor electrode **115b**. In particular, the conductive via **133a** is disposed over the S/D region **121a**, and the conductive layer **135a** is disposed over the conductive via **133a**. The S/D region **121a** is electrically connected to the conductive layer **135a** through the conductive via **133a**, and the conductive layer **135a** is used to electrically connect the S/D region **121a** to other devices.

(40) In some embodiments, the conductive via **133b** is disposed over the S/D region **121b**, the conductive via **133c** is disposed over a portion of the resistor electrode **115b** adjacent to the isolation structure **105b**, and the conductive layer **135b** is disposed over the conductive vias **133b** and **133c**. The conductive vias **133b** and **133c** are covered by the conductive layer **135b**. It should be noted that the S/D region **121b** of the transistor **100** is electrically connected to the resistor electrode **115b** of the resistor **200** through the interconnect structure **137** (i.e., the conductive via **133b**, the conductive layer **135b**, and the conductive via **133c**). Therefore, the transistor **100** and the resistor **200** are connected in series.

(41) In some embodiments, the conductive via **133d** is disposed over a portion of the resistor electrode **115b** adjacent to the isolation structure **105c**, and the conductive layer **135c** is disposed over the conductive via **133d**. The resistor electrode **115b** is electrically connected to the conductive layer **135c** through the conductive via **133d**, and the conductive layer **135c** is used to electrically connect the resistor **200** to other devices.

(42) FIG. **2** is a circuit diagram of the semiconductor device structure **300a** in FIG. **1**. As shown in FIGS. **1** and **2**, the transistor **100** of the semiconductor device structure **300a** may be a P-type metal-oxide-semiconductor (PMOS) transistor **100P**, and the PMOS transistor **100P** and the resistor **200** are connected in series. Alternatively, the transistor **100** of the semiconductor device structure **300a** may be an N-type metal-oxide-semiconductor (NMOS) transistor **100N**, and the NMOS transistor **100N** and the resistor **200** are connected in series.

(43) FIG. **3** is a cross-sectional view illustrating a modified semiconductor device structure **300b**, in accordance with some embodiments. Similar to the semiconductor device structure **300a** of FIG. **1**, the semiconductor device structure **300b** includes the transistor **100** and the resistor **200**. The difference between the semiconductor device structures **300a** and **300b** is the connecting fashion of the transistor **100** and the resistor **200**.

(44) As shown in FIG. 3, the semiconductor device structure **300b** includes a conductive structure **145** disposed in the semiconductor substrate **101** and over the isolation structure **105b**, in accordance with some embodiments. In some embodiments, the conductive structure **145** is covered by the ILD layer **123'**, and the conductive structure **145** is disposed between the S/D region **121b** of the transistor **100** and the resistor electrode **115b** of the resistor **200**. It should be noted that the S/D region **121b** is electrically connected to the resistor electrode **115b** through the conductive structure **145**.

(45) In some embodiments, the conductive structure **145** is in direct contact with the S/D region **121b** and the resistor electrode **115b**, and the conductive structure **145** is not covered by the dielectric layer **113'**. Moreover, in some embodiments, the conductive vias **133b** and **133c**, and the conductive layer **135b** are not formed in the semiconductor device structure **300b**, and the interconnect structure **137** of the semiconductor device structure **300b** includes the conductive vias **133a**, **133d** and the conductive layers **135a**, **135c**. FIG. 2 may be a circuit diagram represent the semiconductor device structure **300b** shown in FIG. 3.

(46) FIG. 4 is a flow diagram illustrating a method **10** of forming a semiconductor device structure (e.g., the semiconductor device structures **300a** and **300b**, the semiconductor device structure **400a** shown in FIG. 25, and the semiconductor device structure **400b** shown in FIG. 29), and the method **10** includes steps **S11**, **S13**, **S15**, **S17**, **S19**, **S21-1**, **S21-2**, **S23-1**, **S23-2** and **S25**, in accordance with some embodiments. The steps **S11** to **S25** of FIG. 4 are elaborated in connection with following figures.

(47) FIGS. 5 to 15 are cross-sectional views illustrating intermediate stages in the formation of the semiconductor device structure **300a**, in accordance with some embodiments.

(48) As shown in FIG. 5, the semiconductor substrate **101** is provided. The semiconductor substrate **101** may be a semiconductor wafer such as a silicon wafer. Alternatively or additionally, the semiconductor substrate **101** may include elementary semiconductor materials, compound semiconductor materials, and/or alloy semiconductor materials. Examples of the elementary semiconductor materials may include, but are not limited to, crystal silicon, polycrystalline silicon, amorphous silicon, germanium, and/or diamond. Examples of the compound semiconductor materials may include, but are not limited to, silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide. Examples of the alloy semiconductor materials may include, but are not limited to, SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP.

(49) In some embodiments, the semiconductor substrate **101** includes an epitaxial layer. For example, the semiconductor substrate **101** has an epitaxial layer overlying a bulk semiconductor. In some embodiments, the semiconductor substrate **101** is a semiconductor-on-insulator substrate which may include a substrate, a buried oxide layer over the substrate, and a semiconductor layer over the buried oxide layer, such as a silicon-on-insulator (SOI) substrate, a silicon germanium-on-insulator (SGOI) substrate, or a germanium-on-insulator (GOI) substrate. Semiconductor-on-insulator substrates can be fabricated using separation by implantation of oxygen (SIMOX), wafer bonding, and/or other suitable methods.

(50) Still referring to FIG. 5, the isolation structures **105a**, **105b**, **105c** are formed in the semiconductor substrate **101**, and the well region **103** is formed between the isolation structures **105b** and **105c**, in accordance with some embodiments. In some embodiments, the isolation structures **105a**, **105b** and **105c** are shallow trench isolation (STI) structures. In addition, the isolation structures **105a**, **105b** and **105c** may be made of silicon oxide, silicon nitride, silicon oxynitride or another applicable dielectric material, and the formation of the isolation structures **105a**, **105b** and **105c** may include forming a patterned mask (not shown) over the semiconductor substrate **101**, etching the semiconductor substrate **101** to form openings (not shown) by using the patterned mask as a mask, depositing a dielectric material in the openings and over the semiconductor substrate **101**, and polishing the dielectric material until the semiconductor substrate

101 is exposed.

(51) In some embodiments, the well region **103** is formed by an ion implantation process, and P-type dopants, such as boron (B), gallium (Ga), or indium (In), or N-type dopants, such as phosphorous (P) or arsenic (As), can be implanted in the portion of the semiconductor substrate **101** between the isolation structures **105b** and **105c** to form the well region **103** (The ion implantation process may be performed by using a patterned mask covering the portion of the semiconductor substrate **101** between the isolation structures **105a** and **105b**). The respective step is illustrated as the step **S11** in the method **10** shown in FIG. 4.

(52) In some embodiments, the isolation structures **105a**, **105b** and **105c** are formed before the well region **103**. In some other embodiments, the well region **103** is formed before the isolation structures **105a**, **105b** and **105c**. In addition, the bottom surface **B1** of the well region **103** is higher than the bottom surface **B2** of the isolation structures **105a**, **105b** and **105c**, in accordance with some embodiments.

(53) Next, a patterned mask **107** is formed over the semiconductor substrate **101**, and the semiconductor substrate **101** is recessed to form openings **110a** and **110b** by using the patterned mask **107** as an etching mask, as shown in FIG. 6 in accordance with some embodiments. In some embodiments, the opening **110a** is located between the isolation structures **105a** and **105b**, and the opening **110b** is located between the isolation structures **105b** and **105b**. The respective step is illustrated as the step **S13** in the method **10** shown in FIG. 4.

(54) The patterned mask **107** may be formed by a deposition process and a patterning process. The deposition process for forming the patterned mask **107** may be a chemical vapor deposition (CVD) process, a high-density plasma CVD (HDPCVD) process, a spin-coating process, or another applicable process. The patterning process for forming the patterned mask **107** may include a photolithography process and an etching process. The photolithography process may include photoresist coating (e.g., spin-coating), soft baking, mask aligning, exposure, post-exposure baking, developing the photoresist, rinsing and drying (e.g., hard baking). The etching process may include a dry etching process or a wet etching process.

(55) After the patterned mask **107** is formed, the portions of the semiconductor substrate **101** exposed by the patterned mask **107** are partially removed by an etching process. The etching process may be a wet etching process, a dry etching process, or a combination thereof. In some embodiments, the opening **110b** is formed by removing an upper portion of the well region **103**, such that the opening **110b** is formed over the resulting well region **103**.

(56) In some embodiments, the sidewalls of the isolation structures **105b** and **105c** are partially exposed by the opening **110b**. For example, the isolation structure **105b** has a first sidewall **SW1** facing the isolation structure **105a** and a second sidewall **SW2** facing the isolation structure **105c**. The first sidewall **SW1** is covered by the semiconductor structure **101** while the second sidewall **SW2** is partially exposed by the opening **110b**. Moreover, the opening **110a** has a width **W1**, the opening **110b** has a width **W2**, and the width **W2** is greater than the width **W1**, in accordance with some embodiments. After the openings **110a** and **110b** are formed, the patterned mask **107** may be removed.

(57) The dielectric layer **113** is deposited over the semiconductor substrate **101**, as shown in FIG. 7 in accordance with some embodiments. In some embodiments, the dielectric layer **113** is deposited conformally in the openings **110a** and **110b**, such as on the sidewalls and the bottom surfaces of the openings **110a** and **110b**, and the isolation structures **105a**, **105b** and **105c** are covered by the dielectric layer **113**. The respective step is illustrated as the step **S15** in the method **10** shown in FIG. 4. After the dielectric layer **113** is formed, reduced openings **110a'** and **110b'** are obtained.

(58) In some embodiments, the dielectric layer **113** includes silicon oxide, silicon nitride, silicon oxynitride, or multilayers thereof. In some embodiments, the dielectric layer **113** is made of a high-k dielectric material, such as hafnium oxide, lanthanum oxide, aluminum oxide, zirconium oxide, or the like. In addition, the dielectric layer **113** may be deposited by a conformal deposition

process, such as a CVD process, an atomic layer deposition (ALD) process, a plasma-enhanced CVD (PECVD) process, another applicable process, or a combination thereof.

(59) After the dielectric layer **113** is formed, an electrode layer **115** is formed over the dielectric layer **113**, and the openings **110a'** and **110b'** (see FIG. 7) in the semiconductor substrate **101** are filled by the electrode layer **115**, as shown in FIG. 8 in accordance with some embodiments. The respective step is illustrated as the step **S17** in the method **10** shown in FIG. 4.

(60) In some embodiments, the electrode layer **115** is made of a semiconductor material such as polysilicon. In some embodiments, the electrode layer **115** is deposited over the dielectric layer **113** using a CVD process, an ALD process, a sputtering process, or one or more other applicable processes.

(61) Next, a patterned mask **117** is formed to cover the active region between the isolation structures **105b** and **105c** (i.e., the active region of the subsequently-formed resistor **200**), and an ion implantation process **160** is performed on the portion of the electrode layer **115** exposed by the patterned mask **117**, as shown in FIG. 9 in accordance with some embodiments. In some embodiments, the portion of the electrode layer **115** over the well region **103** is covered by the patterned mask **117**.

(62) Some processes used to form the patterned mask **117** are similar to, or the same as, those used to form the patterned mask **107** (see FIG. 6), and details thereof are not repeated herein. During the ion implantation process **160**, P-type dopants, such as boron (B), gallium (Ga), or indium (In), or N-type dopants, such as phosphorous (P) or arsenic (As), are introduced into the electrode layer **115** using the patterned mask **117** as an implantation mask. After the ion implantation process **160**, the patterned mask **117** may be removed.

(63) Then, a patterned mask **119** is formed to cover the active region between the isolation structures **105a** and **105b** (i.e., the active region of the subsequently-formed transistor **100**), and an ion implantation process **170** is performed on the portion of the electrode layer **115** exposed by the patterned mask **119**, as shown in FIG. 10 in accordance with some embodiments. In alternative embodiments, the ion implantation process **170** is performed before the ion implantation process **160**. The respective step is illustrated as the step **S19** in the method **10** shown in FIG. 4.

(64) Some processes used to form the patterned mask **119** are similar to, or the same as, those used to form the patterned mask **107** (see FIG. 6), and details thereof are not repeated herein. During the ion implantation process **170**, P-type dopants, such as boron (B), gallium (Ga), or indium (In), or N-type dopants, such as phosphorous (P) or arsenic (As), are introduced into the electrode layer **115** using the patterned mask **119** as an implantation mask.

(65) It should be noted that the portion of the electrode layer **115** between the isolation structures **105a** and **105b** is heavily doped compared with the portion of the electrode layer **115** between the isolation structures **105b** and **105c** for increased conductivity, in accordance with some embodiments. In some embodiments, the dose amount of the ion implantation process **160** is greater than the dose amount of the ion implantation process **170**. After the ion implantation process **170**, the patterned mask **119** may be removed. In addition, an annealing process may be used to activate the implanted dopants.

(66) Subsequently, a polishing process is performed on the electrode layer **115** to form the gate electrode **115a** in the opening **110a'** (see FIG. 7) and the resistor electrode **115b** in the opening **110b'** (see FIG. 7), as shown in FIG. 11 in accordance with some embodiments. In some embodiments, the polishing process is performed until the dielectric layer **113** is exposed, and the excess portion of the electrode layer **115** over the dielectric layer **113** is removed. The respective step is illustrated as the step **S21-1** in the method **10** shown in FIG. 4.

(67) In some embodiments, the polishing process is a chemical mechanical polishing (CMP) process. In some embodiments, the gate electrode **115a** has a width **W3**, the resistor electrode **115b** has a width **W4**, and the width **W4** is greater than the width **W3**. Moreover, the required conductivity of the gate electrode **115a** is higher than the required conductivity of the resistor

electrode **115b**. Therefore, as mentioned above, a dopant concentration of the gate electrode **115a** is greater than a dopant concentration of the resistor electrode **115b**.

(68) After the gate electrode **115a** and the resistor electrode **115b** are formed, the S/D regions **121a** and **121b** are formed in the semiconductor substrate **101** and on opposite sides of the gate electrode **115a**, as shown in FIG. **12** in accordance with some embodiments. The S/D regions **121a** and **121b** may be formed by ion implantation and/or diffusion, and an annealing process, such as a rapid thermal annealing (RTA) process, may be used to activate the implanted dopants. The respective step is illustrated as the step **S23-1** in the method **10** shown in FIG. **4**.

(69) In some embodiments, the S/D regions **121a** and **121b**, and the well region **103** are doped with one or more P-type dopants, such as boron (B), gallium (Ga), or indium (In). In alternative embodiments, the S/D regions **121a** and **121b**, and the well region **103** are doped with one or more N-type dopants, such as phosphorous (P) or arsenic (As).

(70) An ILD layer **123** is formed over the structure of FIG. **12**, as shown in FIG. **13** in accordance with some embodiments. In some embodiments, the ILD layer **123** is made of silicon oxide, silicon nitride, silicon oxynitride, phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), low-k dielectric material, and/or other applicable dielectric materials. In addition, the ILD layer **123** may be formed by a CVD process, a physical vapor deposition (PVD) process, an ALD process, a spin-coating process, or another applicable process.

(71) Next, a patterned mask **125** is formed over the ILD layer **123**, as shown in FIG. **14** in accordance with some embodiments. In some embodiments, the patterned mask **125** has openings, and portions of the ILD layer **123** are exposed by the openings of the patterned mask **125**. Some processes used to form the patterned mask **125** are similar to, or the same as, those used to form the patterned mask **107** (see FIG. **6**), and details thereof are not repeated herein.

(72) Then, an etching process is performed on the ILD layer **123** and the dielectric layer **113** using the patterned mask **125** as a mask, as shown in FIG. **15** in accordance with some embodiments. After the etching process, openings **130a**, **130b**, **130c** and **130d** are formed. In some embodiments, openings **130a** and **130b** penetrate through the ILD layer **123** and the dielectric layer **113**, and the S/D regions **121a** and **121b** are exposed by the openings **130a** and **130b**, respectively. In some embodiments, the openings **130c** and **130d** penetrate through the ILD layer **123**, and the resistor electrode **115b** are partially exposed by the openings **130c** and **130d**. After the openings **130a**, **130b**, **130c** and **130d** are formed, an etched ILD layer **123'** and an etched dielectric layer **113'** are obtained.

(73) Referring back to FIG. **1**, the interconnect structure **137** is formed over the ILD layer **123'**, in accordance with some embodiments. As mentioned above, the interconnect structure **137** includes the conductive vias **133a**, **133b**, **133c** and **133d**, and the conductive layers **135a**, **135b** and **135c**. In some embodiments, the conductive vias **133a**, **133b**, **133c** and **133d** are formed in the openings **130a**, **130b**, **130c** and **130d**, respectively, and the conductive layers **135a**, **135b** and **135c** are formed over the ILD layer **123'** to cover the conductive vias **133a**, **133b**, **133c** and **133d**. The respective step is illustrated as the step **S25** in the method **10** shown in FIG. **4**.

(74) In some embodiments, the interconnect structure **137** is made of copper (Cu), copper alloy, aluminum (Al), aluminum alloy, tungsten (W), tungsten alloy, titanium (Ti), titanium alloy, tantalum (Ta), tantalum alloy, or a combination thereof. Alternatively, other applicable conductive materials may be used. Moreover, the interconnect structure **137** may be formed by one or more deposition process, and a subsequent patterning process. The deposition process may be a CVD process, a PVD process, an ALD process, a metal organic CVD (MOCVD) process, a sputtering process, a plating process, or another applicable deposition process, and the patterning process may include a photolithography process and an etching process. In some embodiments, the interconnect structure **137** includes multilayers.

(75) FIGS. **16** to **17** are cross-sectional views illustrating intermediate stages in the formation of the semiconductor device structure **300a**, in accordance with some other embodiments.

(76) In alternative embodiments, the ion implantation processes **160** and **170** (see FIGS. **9** and **10**) are replaced by ion implantation processes **180** and **190**, as shown in FIGS. **16** and **17** in accordance with some embodiments. As shown in FIG. **16**, the ion implantation process **180** is performed on the structure of FIG. **8** without using any implantation mask.

(77) Subsequently, a patterned mask **139** is formed to cover the active region between the isolation structures **105b** and **105c** (i.e., the active region of the subsequently-formed resistor **200**), and the ion implantation process **190** is performed on the portion of the electrode layer **115** exposed by the patterned mask **139**, as shown in FIG. **17** in accordance with some embodiments. Some processes used to form the patterned mask **139** are similar to, or the same as, those used to form the patterned mask **107** (see FIG. **6**), and details thereof are not repeated herein.

(78) It should be noted that the portion of the electrode layer **115** between the isolation structures **105a** and **105b** are ion-implanted one time more than the portion of the electrode layer **115** between the isolation structures **105b** and **105c**. Therefore, the dopant concentration of the portion of the electrode layer **115** between the isolation structures **105a** and **105b** is greater than the dopant concentration of the portion of the electrode layer **115** between the isolation structures **105b** and **105c**. As a consequence, the dopant concentration of the gate electrode **115a** is greater than a dopant concentration of the resistor electrode **115b** in the resulting semiconductor device structure **300a**.

(79) FIGS. **18** to **20** are cross-sectional views illustrating intermediate stages in the formation of the modified semiconductor device structure **300b**, in accordance with some embodiments.

(80) As shown in FIG. **18**, an upper portion of the isolation structure **105b** is etched to form an opening **140** after the electrode layer **115** is polished (see FIG. **11**), in accordance with some embodiments. The opening **140** may be formed by using a patterned mask (not shown) as an etching mask. In some embodiments, the dielectric layer **113**, the semiconductor substrate **101**, the resistor electrode **115b** are partially etched to form the opening **140**.

(81) Next, a conductive layer **143** is formed over the dielectric layer **113**, and the opening **140** is filled by the conductive layer **143**, as shown in FIG. **19** in accordance with some embodiments. In some embodiments, the conductive layer **143** is in direct contact with the resistor electrode **115b**.

(82) In some embodiments, the conductive layer **143** is made of copper (Cu), copper alloy, aluminum (Al), aluminum alloy, tungsten (W), tungsten alloy, titanium (Ti), titanium alloy, tantalum (Ta), tantalum alloy, another applicable conductive material, or a combination thereof. In addition, the conductive layer **143** may be formed by a CVD process, a PVD process, an ALD process, a plating process, a sputtering process, or another applicable process.

(83) After the conductive layer **143** is formed, a planarization process is performed to expose the gate electrode **115a** and the resistor electrode **115b**, and the conductive structure **145** is formed in the semiconductor substrate **101** and over the isolation structure **105b**, as shown in FIG. **20** in accordance with some embodiments. The planarization process for forming the conductive structure **145** may include a CMP process, a grinding process, an etching process, another suitable process, or a combination thereof.

(84) After the planarization process, the top surfaces of the dielectric layer **113**, the gate electrode **115a**, the resistor electrode **115b**, and the conductive structure **145** are substantially coplanar with each other, in accordance with some embodiments. Within the context of this disclosure, the word “substantially” means preferably at least 90%, more preferably 95%, even more preferably 98%, and most preferably 99%.

(85) Subsequently, the processes of FIGS. **12** to **15** used to form the semiconductor device structure **300a** are performed on the structure of FIG. **20**. As mentioned above, since the connecting fashion of the transistor **100** and the resistor **200** are different between the semiconductor device structures **300a** and **300b**, the openings **130b** and **130c** (see FIG. **15**) are not formed in the ILD layer **123'**, such that the interconnect structure **137** of the semiconductor device structure **300b** (see FIG. **3**) only includes the conductive vias **133a**, **133d**, and the conductive layers **135a**, **135c**. However, in

some other embodiments, the interconnect structure **137** of the semiconductor device structure **300b** includes other conductive elements for electrically connecting with other devices.

(86) FIGS. **21** to **24** are cross-sectional views illustrating intermediate stages in the formation of a semiconductor device structure **400a** shown in FIG. **25**, in accordance with some embodiments. It should be pointed out that operations for forming the semiconductor device structure **400a** before the structure shown in FIG. **21** are substantially the same as the operations for forming the semiconductor device structure **300a** shown in FIGS. **5-7** (The respective steps are illustrated as the steps **S11** to **S15** in the method **10** shown in FIG. **4**), and the related detailed descriptions may refer to the foregoing paragraphs and are not discussed again herein.

(87) After the dielectric layer **113** is formed, an electrode layer **415** is formed over the dielectric layer **113**, as shown in FIG. **21** in accordance with some embodiments. In some embodiments, the openings **110a'** and **110b'** (see FIG. **7**) in the semiconductor substrate **101** are filled by the electrode layer **415**, and an air gap **420** is enclosed (or sealed) in the portion of the electrode layer **415** in the opening **110a'**. In other words, the air gap **420** is not exposed. The respective step is illustrated as the step **S17** in the method **10** shown in FIG. **4**.

(88) In some embodiments, the electrode layer **415** is made of a semiconductor material such as polysilicon. In some embodiments, the electrode layer **415** is deposited over the dielectric layer **113** using a CVD process, an ALD process, a sputtering process, or one or more other applicable processes.

(89) Next, similar to the operation shown in FIG. **9**, a patterned mask **117** is formed to cover the active region between the isolation structures **105b** and **105c** (i.e., the active region of the subsequently-formed resistor **200**), and an ion implantation process **160** is performed on the portion of the electrode layer **415** exposed by the patterned mask **117**, as shown in FIG. **22** in accordance with some embodiments. In some embodiments, the portion of the electrode layer **415** over the well region **103** is covered by the patterned mask **117**.

(90) Some processes used to form the patterned mask **117** in FIG. **22** are similar to, or the same as, those used to form the patterned mask **107** in FIG. **6**, and details thereof are not repeated herein. During the ion implantation process **160**, P-type dopants, such as boron (B), gallium (Ga), or indium (In), or N-type dopants, such as phosphorous (P) or arsenic (As), are introduced into the electrode layer **415** using the patterned mask **117** as an implantation mask. After the ion implantation process **160**, the patterned mask **117** may be removed.

(91) Then, similar to the operation shown in FIG. **10**, a patterned mask **119** is formed to cover the active region between the isolation structures **105a** and **105b**, and an ion implantation process **170** is performed on the portion of the electrode layer **415** exposed by the patterned mask **119**, as shown in FIG. **23** in accordance with some embodiments. In alternative embodiments, the ion implantation process **170** is performed before the ion implantation process **160**. The respective step is illustrated as the step **S19** in the method **10** shown in FIG. **4**.

(92) Some processes used to form the patterned mask **119** in FIG. **23** are similar to, or the same as, those used to form the patterned mask **107** in FIG. **6**, and details thereof are not repeated herein. During the ion implantation process **170**, P-type dopants, such as boron (B), gallium (Ga), or indium (In), or N-type dopants, such as phosphorous (P) or arsenic (As), are introduced into the electrode layer **415** using the patterned mask **119** as an implantation mask.

(93) It should be noted that the portion of the electrode layer **415** between the isolation structures **105a** and **105b** is heavily doped compared with the portion of the electrode layer **415** between the isolation structures **105b** and **105c** for increased conductivity, in accordance with some embodiments. In some embodiments, the dose amount of the ion implantation process **160** is greater than the dose amount of the ion implantation process **170**.

(94) After the ion implantation process **170**, the patterned mask **119** may be removed. In addition, an annealing process may be used to activate the implanted dopants. In alternative embodiments, the ion implantation processes **160** and **170** shown in FIGS. **22** and **23** are replaced by the ion

implantation processes **180** and **190** (see FIGS. **16** and **17**).

(95) Subsequently, a polishing process is performed on the electrode layer **415** to form a fuse **415a** in the opening **110a'** (see FIG. **7**) and a resistor electrode **415b** in the opening **110b'** (see FIG. **7**), as shown in FIG. **24** in accordance with some embodiments. In some embodiments, the polishing process is performed until the dielectric layer **113** is exposed, and the excess portion of the electrode layer **415** over the dielectric layer **113** is removed. The respective step is illustrated as the step **S21-2** in the method **10** shown in FIG. **4**.

(96) In some embodiments, the polishing process is a CMP process. In some embodiments, the isolation structure **105b** is closer to the resistor electrode **415b** than the fuse **415a**. In some embodiments, the air gap **420** is enclosed (or sealed) in the fuse **415a**. In some embodiments, the fuse **415a** is an electrical fuse (e-fuse). In some embodiments, the fuse **415a** has a top width **W5**, the resistor electrode **415b** has a top width **W6**, and the top width **W6** is greater than the top width **W5**.

(97) In some embodiments, a material of the fuse **415a** and a material of the resistor electrode **415b** are the same, such as polysilicon. However, in some embodiments, a dopant concentration of the fuse **415a** is different from a dopant concentration of the resistor electrode **415b**. For example, in some embodiments, a dopant concentration of the fuse **415a** is greater than a dopant concentration of the resistor electrode **415b**. In some embodiments, a top surface **T1** of the fuse **415a** is substantially coplanar with a top surface **T2** of the resistor electrode **415b**.

(98) After the fuse **415a** and the resistor electrode **415b** are formed, similar to the operations shown in FIGS. **12** to **15**, S/D regions **121a** and **121b** are formed in the semiconductor substrate **101** and on opposite sides of the fuse **415a**, and an ILD layer **123'** and an interconnect structure **137** are formed over the dielectric layer **113'**, as shown in FIG. **25** in accordance with some embodiments. The respective steps are illustrated as the steps **S23-2** and **S25** in the method **10** shown in FIG. **4**.

(99) In some embodiments, the interconnect structure **137** includes conductive vias **133a**, **133b**, **133c** and **133d**, and conductive layers **135a**, **135b** and **135c**. The details (e.g., the materials and processes) of the S/D regions **121a** and **121b**, the ILD layer **123'** and the interconnect structure **137** may be essentially the same as what are shown and discussed in FIG. **1** and FIGS. **12** to **15**, and hence are not repeated herein. After the interconnect structure **137** is formed, the semiconductor device structure **400a** is obtained.

(100) In some embodiments, the semiconductor device structure **400a** includes the fuse **415a** and a resistor **200**. As mentioned above, the top surface **T1** of the fuse **415a** is substantially coplanar with the top surface **T2** of the resistor electrode **415b** of the resistor **200**, in accordance with some embodiments. Moreover, the isolation structure **105b** between the fuse **415a** and the resistor electrode **415b** has a top surface **T3** covered by and in direct contact with the dielectric layer **113'**.

(101) In some embodiments, the second sidewall **SW2** of the isolation structure **105b** is in direct contact with the dielectric layer **113'**. In some embodiments, the dielectric layer **113'** has a first portion **113'a** extending between the fuse **415a** and the semiconductor substrate **101**, and a second portion **113'b** extends between the resistor electrode **415b** and the well region **103**. In addition, the S/D region **121b** is electrically connected to the resistor electrode **415b** of the resistor **200** through the conductive via **133b**, the conductive layer **135b** and the conductive via **133c** of the interconnect structure **137**, in accordance with some embodiments.

(102) FIGS. **26** to **28** are cross-sectional views illustrating intermediate stages in the formation of a semiconductor device structure **400b** shown in FIG. **29**, in accordance with some embodiments. It should be pointed out that operations for forming the semiconductor device structure **400b** before the structure shown in FIG. **26** are substantially the same as the operations for forming the semiconductor device structure **300a** shown in FIGS. **5-7** and the subsequent operations for forming the semiconductor device structure **400a** shown in FIGS. **21-24**. (The respective steps are illustrated as the steps **S11** to **S21-2** in the method **10** shown in FIG. **4**), and the related detailed descriptions may refer to the foregoing paragraphs and are not discussed again herein.

(103) After the electrode layer **415** is polished to form the fuse **415a** and the resistor electrode **415b** (see FIG. **24**), similar to the operation shown in FIG. **18**, an upper portion of the isolation structure **105b** is etched to form an opening **140**, in accordance with some embodiments. The opening **140** may be formed by using a patterned mask (not shown) as an etching mask. In some embodiments, the dielectric layer **113**, the semiconductor substrate **101**, the resistor electrode **415b** are partially etched to form the opening **140**.

(104) Next, similar to the operation shown in FIG. **19**, a conductive layer **143** is formed over the dielectric layer **113**, and the opening **140** is filled by the conductive layer **143**, as shown in FIG. **27** in accordance with some embodiments. In some embodiments, the conductive layer **143** is in direct contact with the resistor electrode **415b**. The details (e.g., the materials and processes) of the conductive layer **143** may be essentially the same as what are shown and discussed in FIG. **19**, and hence are not repeated herein.

(105) After the conductive layer **143** is formed, similar to the operation shown in FIG. **20**, a planarization process is performed to expose the fuse **415a** and the resistor electrode **415b**, and a conductive structure **145** is formed in the semiconductor substrate **101** and over the isolation structure **105b**, as shown in FIG. **28** in accordance with some embodiments. The planarization process for forming the conductive structure **145** may include a CMP process, a grinding process, an etching process, another suitable process, or a combination thereof. After the planarization process, the top surface T1 of the fuse **415a**, the top surface T2 of the resistor electrode **415b** and the top surface T4 of the conductive structure **145** are substantially coplanar with each other, in accordance with some embodiments.

(106) Subsequently, the processes of FIGS. **12** to **15** used to form the semiconductor device structure **300a** are performed on the structure of FIG. **28**, such that the semiconductor device structure **400b** is obtained, as shown in FIG. **29** in accordance with some embodiments. In some embodiments, the semiconductor device structure **400b** includes the fuse **415a** and a resistor **200**. Since the connecting fashion of the fuse **415a** and the resistor **200** are different between the semiconductor device structures **400a** and **400b**, the openings **130b** and **130c** (see FIG. **15**) are not formed in the ILD layer **123'**, such that the interconnect structure **137** of the semiconductor device structure **400b** (see FIG. **29**) only includes the conductive vias **133a**, **133d**, and the conductive layers **135a**, **135c**. However, in some other embodiments, the interconnect structure **137** of the semiconductor device structure **400b** includes other conductive elements for electrically connecting with other devices.

(107) In some embodiments, the top surface T1 of the fuse **415a**, the top surface T2 of the resistor electrode **415b** of the resistor **200**, and the top surface T4 of the conductive structure **145** are substantially coplanar with each other. In some embodiments, the second sidewall SW2 of the isolation structure **105b** is in direct contact with the dielectric layer **113'**. In some embodiments, the dielectric layer **113'** has a first portion **113'a** extending between the fuse **415a** and the semiconductor substrate **101**, and a second portion **113'b** extends between the resistor electrode **415b** and the well region **103**. In addition, the S/D region **121b** is electrically connected to the resistor electrode **415b** of the resistor **200** through the conductive structure **145**, in accordance with some embodiments.

(108) Embodiments of the semiconductor device structures **300a** and **300b** are provided in the disclosure. The semiconductor device structures **300a** and **300b** include the transistor **100** (e.g., the PMOS transistor **100P** or the NMOS transistor **100N**) and the resistor **200** connected in series and formed by an integrated process flow. Particularly, the gate electrode **115a** of the transistor **100** and the resistor electrode **115b** of the resistor **200** are formed in the semiconductor substrate **101** by the same process steps. Therefore, the resistor **200** may have high sheet resistance without using additional masks or process steps. As a result, the associated costs may be reduced, and the performance of the semiconductor device structures **300a** and **300b** may be improved.

(109) Moreover, embodiments of the semiconductor device structures **400a** and **400b** are provided

in the disclosure. The semiconductor device structures **400a** and **400b** include the fuse **415a** and the resistor **200** connected in series and formed by an integrated process flow. Since the fuse **415a** and the resistor electrode **415b** of the resistor **200** are formed in the semiconductor substrate **101** by the same process steps, the associated costs may be reduced, and the performance of the semiconductor device structures **400a** and **400b** may be improved.

(110) In one embodiment of the present disclosure, a semiconductor device structure is provided. The semiconductor device structure includes an isolation structure disposed in a semiconductor substrate. The semiconductor device structure also includes a fuse and a resistor electrode disposed in the semiconductor substrate. The isolation structure is disposed between the fuse and the resistor electrode, and the isolation structure is closer to the resistor electrode than the fuse. The semiconductor device structure further includes a source/drain (S/D) region disposed in the semiconductor substrate and between the fuse and the isolation structure. The S/D region is electrically connected to the resistor electrode.

(111) In another embodiment of the present disclosure, a semiconductor device structure is provided. The semiconductor device structure includes a fuse and a resistor electrode disposed in a semiconductor substrate. A dopant concentration of the fuse is greater than a dopant concentration of the resistor electrode. The semiconductor device structure also includes an isolation structure disposed in the semiconductor substrate and between the fuse and the resistor electrode. The semiconductor device structure further includes a source/drain (S/D) region disposed in the semiconductor substrate and between the fuse and the isolation structure. The S/D region is electrically connected to the resistor electrode.

(112) In yet another embodiment of the present disclosure, a method for preparing a semiconductor device structure is provided. The method includes forming an isolation structure in a semiconductor substrate, and recessing the semiconductor substrate to form a first opening and a second opening. The first opening and the second opening are on opposite sides of the isolation structure, and a width of the second opening is greater than a width of the first opening. The method also includes forming an electrode layer over the semiconductor substrate, and polishing the electrode layer to form a fuse in the first opening and a resistor electrode in the second opening. The method further includes forming a source/drain (S/D) region in the semiconductor substrate. The S/D region is between the fuse and the isolation structure.

(113) The embodiments of the present disclosure have some advantageous features. By forming a gate electrode of a transistor and a resistor electrode of a resistor in a semiconductor substrate by the same process steps, the resistor may have high sheet resistance without using additional masks or process steps. This significantly reduces the associated costs, and the performance of the semiconductor device structure including the transistor and the resistor may be improved.

(114) Although the present disclosure and its advantages have been described in detail, it should be understood that various changes, substitutions and alterations can be made herein without departing from the spirit and scope of the disclosure as defined by the appended claims. For example, many of the processes discussed above can be implemented in different methodologies and replaced by other processes, or a combination thereof.

(115) Moreover, the scope of the present application is not intended to be limited to the particular embodiments of the process, machine, manufacture, composition of matter, means, methods and steps described in the specification. As one of ordinary skill in the art will readily appreciate from the disclosure of the present invention, processes, machines, manufacture, compositions of matter, means, methods, or steps, presently existing or later to be developed, that perform substantially the same function or achieve substantially the same result as the corresponding embodiments described herein may be utilized according to the present disclosure. Accordingly, the appended claims are intended to include within their scope such processes, machines, manufacture, compositions of matter, means, methods, and steps.

Claims

1. A semiconductor device structure, comprising: an isolation structure disposed in a semiconductor substrate; a fuse and a resistor electrode disposed in the semiconductor substrate, wherein the isolation structure is disposed between the fuse and the resistor electrode, and the isolation structure is closer to the resistor electrode than the fuse; and a source/drain (S/D) region disposed in the semiconductor substrate and between the fuse and the isolation structure, wherein the S/D region is electrically connected to the resistor electrode.
2. The semiconductor device structure of claim 1, wherein an air gap is enclosed in the fuse.
3. The semiconductor device structure of claim 1, wherein the fuse is an electrical fuse.
4. The semiconductor device structure of claim 1, wherein a top width of the resistor electrode is greater than a top width of the fuse.
5. The semiconductor device structure of claim 1, wherein the fuse and the resistor electrode comprise polysilicon; wherein a dopant concentration of the fuse is greater than a dopant concentration of the resistor electrode.
6. The semiconductor device structure of claim 1, wherein a top surface of the fuse is substantially coplanar with a top surface of the resistor electrode.
7. The semiconductor device structure of claim 1, further comprising: a well region disposed in the semiconductor substrate, wherein the resistor electrode is disposed over the well region; wherein the well region adjoins the isolation structure.
8. The semiconductor device structure of claim 1, further comprising: a dielectric layer disposed over the semiconductor substrate, wherein a first portion of the dielectric layer extends between the fuse and the semiconductor substrate.
9. The semiconductor device structure of claim 8, wherein a second portion of the dielectric layer extends between the resistor electrode and the semiconductor substrate; wherein the isolation structure is covered by the dielectric layer.
10. The semiconductor device structure of claim 1, further comprising: an interlayer-dielectric (ILD) layer disposed over the semiconductor substrate, wherein the fuse and the resistor electrode are covered by the ILD layer; and an interconnect structure disposed in the ILD layer, wherein the S/D region is electrically connected to the resistor electrode through the interconnect structure.
11. The semiconductor device structure of claim 10, wherein the interconnect structure comprises: a first conductive via and a second conductive via disposed in the ILD layer, wherein the first conductive via is electrically connected to the S/D region, and the second conductive via is electrically connected to the resistor electrode; and a conductive layer disposed over the ILD layer and electrically connecting the first conductive via and the second conductive via.
12. The semiconductor device structure of claim 1, further comprising: a conductive structure disposed in the semiconductor substrate and over the isolation structure, wherein the S/D region is electrically connected to the resistor electrode through the conductive structure.
13. A semiconductor device structure, comprising: a fuse and a resistor electrode disposed in a semiconductor substrate, wherein a dopant concentration of the fuse is greater than a dopant concentration of the resistor electrode; an isolation structure disposed in the semiconductor substrate and between the fuse and the resistor electrode; and a source/drain (S/D) region disposed in the semiconductor substrate and between the fuse and the isolation structure, wherein the S/D region is electrically connected to the resistor electrode.
14. The semiconductor device structure of claim 13, wherein an air gap is enclosed in the fuse.
15. The semiconductor device structure of claim 13, wherein a top width of the resistor electrode is greater than a top width of the fuse; wherein the fuse and the resistor electrode comprise polysilicon; wherein the isolation structure is closer to the resistor electrode than the fuse.
16. The semiconductor device structure of claim 13, further comprising: a well region disposed in

- the semiconductor substrate, wherein the resistor electrode is disposed over the well region, and a conductivity type of the well region is the same as a conductivity type of the S/D region.
17. The semiconductor device structure of claim 16, further comprising: a dielectric layer disposed over the semiconductor substrate, wherein the fuse is separated from the semiconductor substrate by a first portion of the dielectric layer, and the resistor electrode is separated from the well region by a second portion of the dielectric layer.
18. The semiconductor device structure of claim 17, wherein a top surface and a sidewall of the isolation structure are in direct contact with the dielectric layer.
19. The semiconductor device structure of claim 13, further comprising: a conductive structure disposed in the semiconductor substrate and over the isolation structure, wherein the conductive structure is in direct contact with the S/D region and the resistor electrode.
20. The semiconductor device structure of claim 19, wherein a top surface of the fuse, a top surface of the resistor electrode, and a top surface of the conductive structure are substantially coplanar with each other.
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